COLUMBIA UNIVERSITY
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STAT 5703

Statistical Inference and Modeling

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Motivation

- Until now, all models assumed i.i.d. random sample framework
- We now consider a dependence structure between the observations $X_1, \ldots, X_n$
- This can be done with two approaches: *Markov chains* or time series
- Markov models provide a very rich set of tools for handling dependence

Prof. Andrei A. Markov (1856-1922) , published his result in 1906.
Markov later used his theory to study the distribution of vowels in Onegin, written by Pushkin.

Data Science Example

- A common application of Markov chains in data science is text prediction.
- It's an area of NLP (Natural Language Processing) that is commonly used in the tech industry by companies like Google, LinkedIn and Instagram. When you're writing emails, Google predicts and suggests words or phrases to *autocomplete* your email. And when you receive messages on Instagram or LinkedIn, those apps suggest potential replies.
- These are not the applications of a Markov chain we will explore. The types of models these large-scale companies use in production for these features are more complicated. We will stick to simpler examples.

Example: the taxi problem

A taxi company has divided the city into three regions – Northside, Downtown, and Southside. By keeping track of pickups and drop-offs, the company has found that:

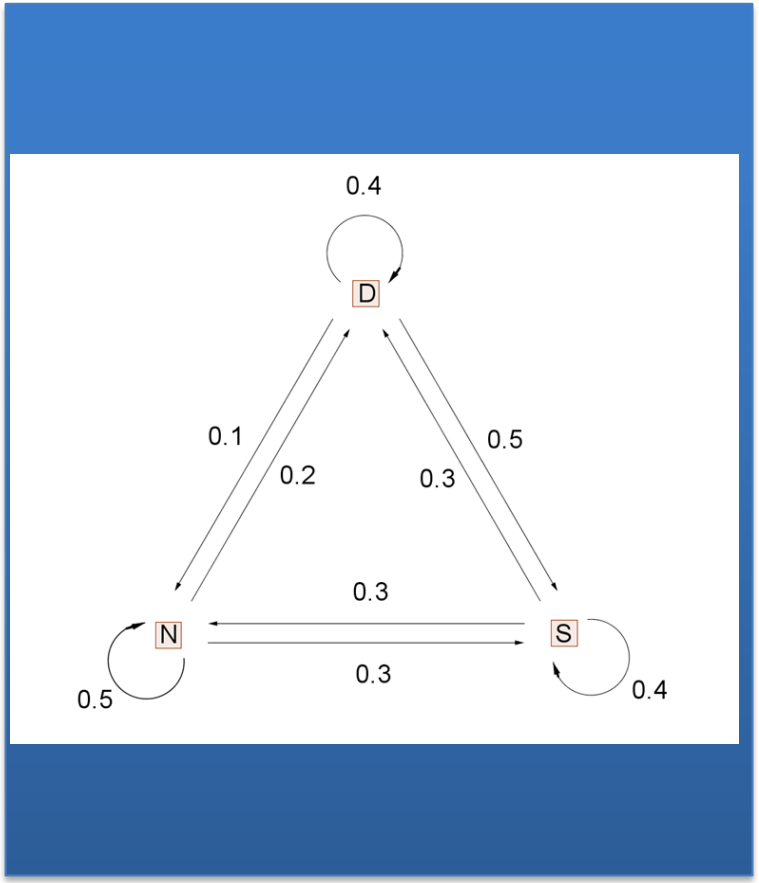
- Of the fares picked up in Northside, 50% stay in that region, 20% are taken to Downtown, and 30% go to Southside.
- Of the fares picked up Downtown, only 10% go to Northside, 40% stay in Downtown, and 50% go to Southside.
- Of the fares picked up in Southside, 30% go to each of Northside and Downtown, while 40% stay in Southside.

We would like to know what the distribution of taxis will be over time as they pick up and drop off successive fares. Suppose we want to know the probability that a taxi starting off Downtown, will be Downtown after letting off its seventh fare?

State Diagram

This information can be represented in a state diagram which includes

- the three states D, N, and S corresponding to the three regions of the city
- the probabilities of a taxi transitioning from one region/state to another



Markov chains

- If the location of the taxi at time *n* is denoted by X_n , then the sequence X_1, X_2, … consists of *dependent* variables, and can be modeled with a Markov chain. The values the random variable X_n can take, are known as *states* of the chain.

- The probabilities of moving from state to state are constant and independent of the past behavior – this property of the system is called the Markov property. That is,

$$P(X_n = s_n | X_{n-1} = s_{n-1}, \dots, X_0 = s_0) = P(X_n = s_n | X_{n-1} = s_{n-1})$$

- We assume that a transition – picking up and dropping off a fare – occurs each time the system is observed, and that observations occur at regular intervals. Systems with these characteristics are called Markov chains or Markov processes (when time is continuous).

Computing Transition Probabilities

- What is the probability that a taxi that starts off Downtown ends up in Northside after two fares?
- One possibility is that the taxi stays Downtown for the first fare and then transitions to Northside for the second. The probability of this occurring is then:

$$\boxed{D} \xrightarrow{0.4} \boxed{D} \xrightarrow{0.1} \boxed{N} \qquad = (0.4)(0.1) = 0.04$$

- But we could also have the taxi going to either Northside of Southside first, then transitioning to Northside:

$$\boxed{D} \xrightarrow{0.5} \boxed{S} \xrightarrow{0.3} \boxed{N} \qquad = (0.5)(0.3) = 0.15$$

$$\boxed{D} \xrightarrow{0.1} \boxed{N} \xrightarrow{0.5} \boxed{N} \qquad = (0.1)(0.5) = 0.05$$

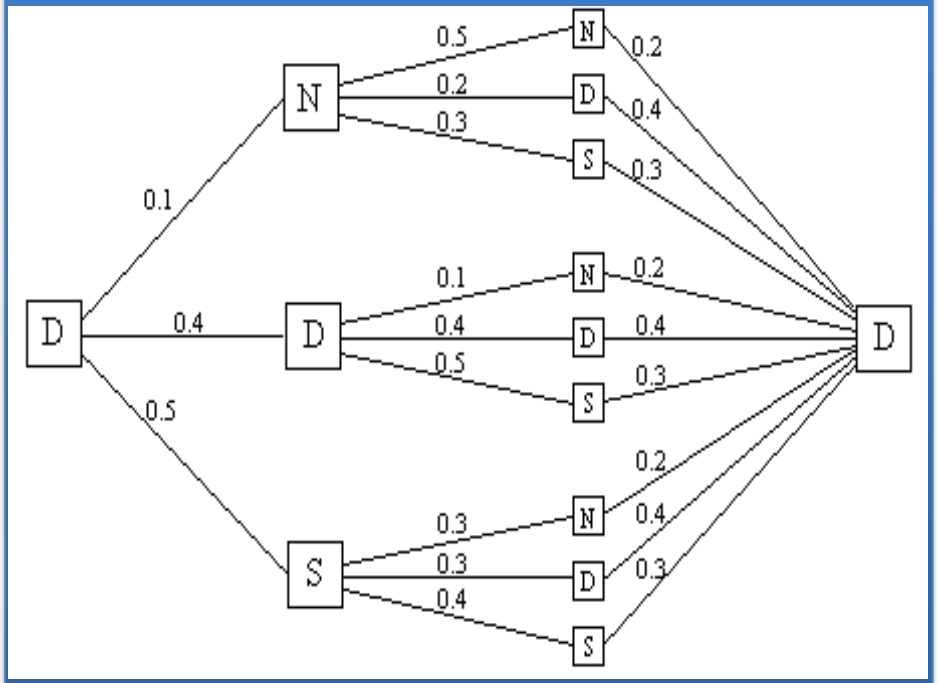
- Since the taxi could follow the first, second or third path, the probability of starting and ending Downtown after two fares is:

$$(0.4)(0.1) \quad + \quad (0.5)(0.3) \quad + \quad (0.1)(0.5) \quad = 0.24$$

$$D \rightarrow D \rightarrow N \quad \underline{\text{or}} \quad D \rightarrow S \rightarrow N \quad \underline{\text{or}} \quad D \rightarrow N \rightarrow N$$

Transitions in More Steps

Tree Diagram



If we multiply along all the paths and sum the results, we find that this probability is 0.309.

- If we want to know the probability of a taxi transitioning from one region to another after just three fares, the computation will have more possible paths.
- Suppose we were interested in the probability of a taxi both starting and ending up Downtown. We can use a tree diagram to represent this calculation.
- More generally, we might want to determine the probability of moving from state *I* to state I over *m* steps.

Transition Matrix

- We can create a square matrix, P, called the **transition matrix,** by constructing rows for the probabilities going from Southside and Northside as well.

To

From

$$\begin{array}{c c} & \begin{array}{ccc} D & S & N \end{array} \\ \begin{array}{c} D \\ S \\ N \end{array} & \begin{bmatrix} 0.4 & 0.5 & 0.1 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{bmatrix} \end{array}$$

- An entry P_{ij} of this matrix is the probability of a transition from region i to region j. For example, p_{32}, is the probability of a fare that originates in Northside going to Southside. (Note the sum of entries across rows of P.)

Transition Matrix

What results when we multiply the transition matrix by itself?

$$P^2 = \begin{bmatrix} 0.4 & 0.5 & 0.1 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.5 & 0.5 \end{bmatrix}^2$$

$$= \begin{bmatrix} 0.4 & 0.5 & 0.1 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.5 & 0.5 \end{bmatrix} \times \begin{bmatrix} 0.4 & 0.5 & 0.1 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.5 & 0.5 \end{bmatrix} = \begin{bmatrix} 0.33 & 0.43 & 0.24 \\ 0.3 & 0.4 & 0.3 \\ 0.27 & 0.37 & 0.36 \end{bmatrix}$$

The highlighted entry results from the same computation that we already considered for of a taxi going from D to N in two fares.

What are the other entries of P^2? What are the entries of P^3? P^n?

Statistical Model

- A Markov chain is a stochastic process $\{X_t\}$ taking values in a (finite discrete) state space $\mathcal{X} = \{1, \ldots, s\}$ such that the distribution of X_t depends only on $X_{t\text{-}1}$. In the taxi example, $\mathcal{X} = \{N, D, S\}$. Note that the state space can also be infinite.

- The observed sample data $X_{t_1}, \ldots, X_{t_n}$ are of the form $X_0 = s_0,\ X_{t_1} = s_1, \ldots,\ X_{t_n} = s_n$, where $0 < t_1 < \ldots < t_n$.

- Note that we can *always* write the likelihood (joint pdf) for any sample, using the multiplication rule for dependent r.v, as:

$$P\left(X_0 = s_0, \ldots, X_{t_n} = s_n\right)$$
$$= P(X_0 = s_0) \prod_{i=1}^{n} P\left(X_{t_i} = s_{t_i} \middle| X_0 = s_0, X_{t_1} = s_{t_1}, \ldots, X_{t_{i-1}} = s_{i-1}\right)$$

- What makes $\{X_t\}$ a Markov chain is the extra assumption that each $P\left(X_{t_i} = s_{t_i} \middle| X_0 = s_0, X_{t_1} = s_{t_1}, \ldots, X_{t_{i-1}} = s_{i-1}\right)$ depends only on the most recent observation $X_{t_{i-1}} = s_{i-1}$

Statistical Model

Definition: The (first-order) *Markov property* assumes that "given the present, the future is independent of the past", meaning that

$$P\left(X_{t_i} = s_{t_i} \middle| X_0 = s_0, X_{t_1} = s_{t_1}, \dots, X_{t_{i-1}} = s_{i-1}\right)$$
$$= P\left(X_{t_i} = s_{t_i} \middle| X_{t_{i-1}} = s_{i-1}\right), \forall i$$

This simplifies the likelihood to:

$$P\left(X_0 = s_0, \dots, X_{t_n} = s_n\right) = P(X_0 = s_0) \prod_{i=1}^{n} P\left(X_{t_i} = s_{t_i} \middle| X_{t_{i-1}} = s_{i-1}\right)$$

For example,

$$P(X_0 = a, X_1 = b, X_2 = c) = P(X_0 = a)P(X_1 = b|X_0 = a)P(X_2 = c|X_1 = b)$$

If the process does not posses the Markov property, then we can only write:

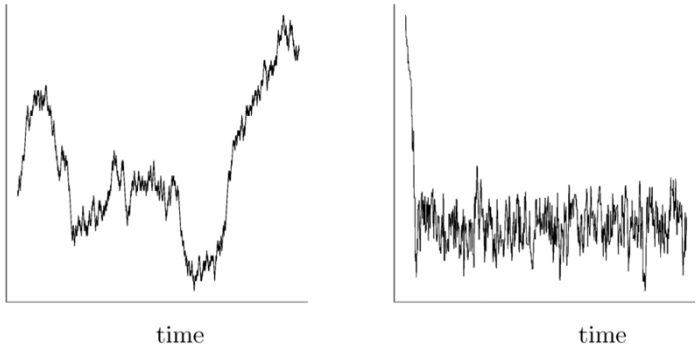
$$P(X_0 = a, X_1 = b, X_2 = c)$$
$$= P(X_0 = a)P(X_1 = b|X_0 = a)P(X_2 = c|X_0 = a, X_1 = b)$$

The theory of Markov chains is a very rich and complex. We must get through many definitions before we can do anything interesting.

Questions to be answered

- When does a Markov chain "settle down" into some sort of equilibrium?
- What does this even mean?
- How do we estimate the parameters of a Markov chain?
- What are the parameters of a Markov chain?
- How can we construct Markov chains that converge to a given equilibrium distribution?
- Why would we want to do that??
- Time to think: do you know what does the following sequence converge to?

$$x_{n+1} = \frac{1}{2}x_n + \frac{1}{x_n}$$



Two Markov chains. One of the chains does not settle down into an equilibrium. The other one does. Which one is which?

Transition Matrix

Definition: A *Markov chain* is a Markov process in discrete time, so we can simply write $t_i = i$, $i = 1, \ldots, n$.

Definition: A *transition matrix* P_t for a Markov chain $\{X\}$ at time t is a matrix containing information on the probability of transitioning between states. Given an ordering of a matrix's rows and columns by the state space $\mathcal{X}$, the $(i, j)^{\text{th}}$ element of the matrix $\boldsymbol{P}_t$ is given by:

$$[\boldsymbol{P}_t]_{ij} = P(X_{t+1} = j | X_t = i)$$

This means each row of the matrix is a *probability vector*, and the sum of its entries is 1.

Transition matrices have the property that the product of subsequent ones describes a transition along the time interval spanned by the transition matrices.

Note*:* The matrix depends on the time t.

Time homogeneous Markov chains

Definition: A Markov chain is *time-independent*, aka *homogeneous* if

$$P(X_{t+1} = s|X_t = r) = P(X_1 = s|X_0 = r), = p_{rs} \; \forall t$$

That is, the conditional probabilities only depend on the time difference.

Note: p_{rs} does **not** depend on t.

For a homogeneous Markov chain X_t, observed at a discrete equally spaced times, $t = 0, 1, \ldots, n$, we define **the** constant $s \times s$ *transition matrix* $\boldsymbol{P}$ such that the $(i, j)^{\text{th}}$ element p_{ij} is given by:

$$[\boldsymbol{P}]_{ij} = P(X_{t+1} = j|X_t = i) = P(X_1 = j|X_0 = i), \forall t$$

Note: The transition matrix must satisfy that

1. $p_{ij} \geq 0, \forall i, j \in \mathcal{X}$
2. $\sum_{j=1}^{s} p_{ij} = 1, \forall i \in \mathcal{X}$

Transition Matrix Properties

Denote the *initial* probabilities by $p_j = P(X_0 = j), j = 1, \dots, s$. That is, we can form a *vector* $\boldsymbol{p}$ of all initial probabilities: $\boldsymbol{p} = \begin{pmatrix} p_1 \\ \vdots \\ p_s \end{pmatrix}$.

Then, by the law of total probability,

$$P(X_1 = k) = \sum_{j=1}^{s} P(X_0 = j) P(X_1 = k | X_0 = j) = \sum_{j=1}^{s} p_j p_{jk}$$

Notice this is the kth element of $\boldsymbol{p}'\boldsymbol{P}$

Theorem: By induction, we can prove that the pmf of X_n is given by $\boldsymbol{p}'\boldsymbol{P}^n$. That is, $P(X_n = k)$ is the kth element of the vector $\boldsymbol{p}'\boldsymbol{P}^n$.

Note: This is different than $\boldsymbol{P}^n$. The element on i^{th} row, j^{th} column of $\boldsymbol{P}^n$ is equal to $P(X_n = j | X_1 = i)$.

Example: Given the chain is in state s, the probability of a *run* of $m \geq 1$ successive stays in state s is $p_{ss}^{m-1}(1 - p_{ss}) \sim \text{Geo}(1 - p_{ss})$

Stationary Distribution

To understand the long-term behavior of our Markov chain, we introduce the concept of a stationary distribution.

Definition: A Markov chain with transition matrix **P** is said to have a *stationary distribution* $\boldsymbol{\pi}$ (aka equilibrium or invariant distribution) if there exists a vector $\boldsymbol{\pi}$ such that $\boldsymbol{\pi}'\boldsymbol{P} = \boldsymbol{\pi}'$

This equation implies that multiplying the equilibrium vector by the transition matrix P yields the same vector. In Markov chains, this is called **"Pisa state"** and it refers to a specific kind of state where the chain can only reach from other states but cannot transition away from it once entered.

Notice this means that $\boldsymbol{\pi}$ must be a left eigenvector, corresponding to eigenvalue 1. Note also that all transition matrices posses a right eigenvector, corresponding to eigenvalue 1, since $\boldsymbol{P}\mathbf{1}_S = \mathbf{1}_S$, where $\mathbf{1}_S$ is a vector of *S* 1's.

Theorem: All Markov chains defined on *finite* state spaces always posses at least one equilibrium distribution.

Theorem: Under some conditions (the Markov chain is ergodic), the stationary distribution satisfies:

$$\lim_{n\to\infty} \boldsymbol{P}^n = \mathbf{1}'\boldsymbol{\pi}$$

Irreducibility

Definition: A state s_j is said to be *accessible* from state s_i if a chain starting in state s_i has a positive probability to reach state s_j at some future time point n. That is,

$$\exists n > 0 : p_{ij}^{n} > 0$$

If s_j is accessible from s_i and s_i is accessible from state s_j then we say that s_i and s_j *communicate*.

A *communicating class* is defined to be a set of states that communicate.

Definition: If a discrete time Markov chain is composed by only *one* communicating class (i.e., if all states in the chain communicate), then it is said to be *irreducible*.

Example: $P = \begin{bmatrix} 0.5 & 0.5 & 0 & 0 \\ 0.9 & 0.1 & 0 & 0 \\ 0 & 0 & 0.2 & 0.8 \\ 0 & 0 & 0.7 & 0.3 \end{bmatrix}$ is reducible and has 2 classes.

Note: Irreducibility does *not* guarantee the presence of limiting probabilities.

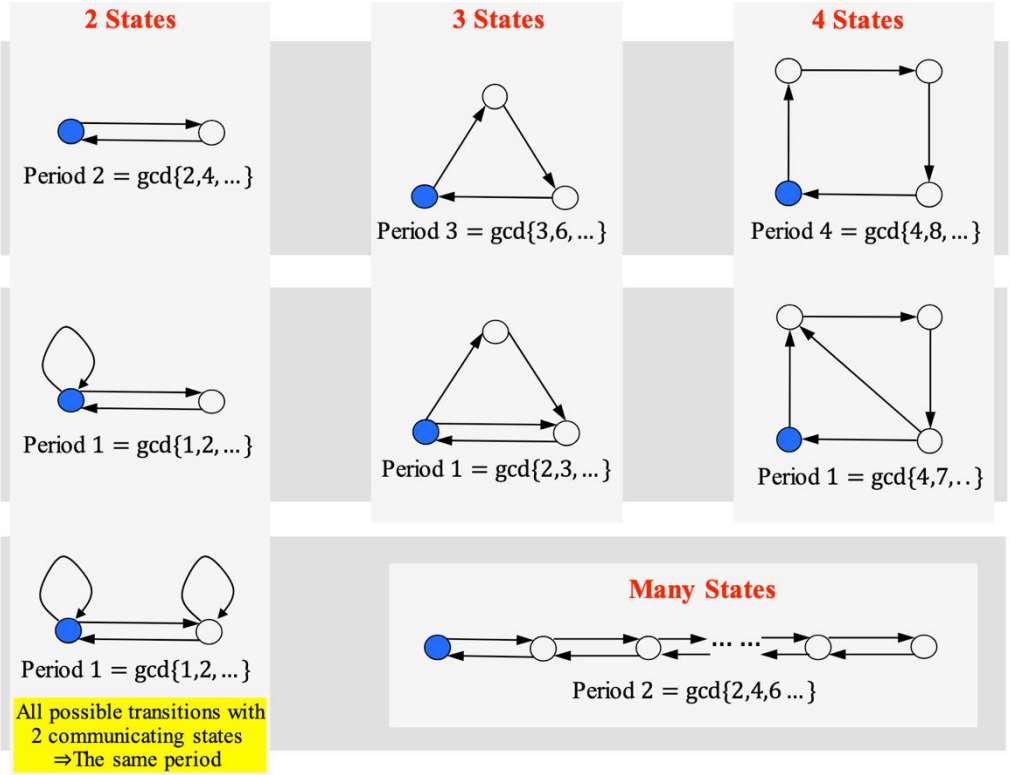
Periodicity

Definition: We define the *period* of state *i,* denoted *d*(*i*), to be the greatest common divisor of

$$J_i = \{n \geq 0 : p_n(i,i) > 0\}$$

That is, any return to state *i* must occur in multiples of *d*(*i*) time steps.

We call an irreducible transition matrix *P* *aperiodic* if $d(i) = 1, \forall i$.



Periodicity

***Theorem**:* If a stochastic matrix is irreducible and aperiodic, then there exists a *unique* invariant distribution and it satisfies the ergodic theorem.

Examples:

$P = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ has infinitely many invariant distributions.

$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ has a unique invariant distribution but is not ergodic.

Which condition is violated in each case?

***Exercise**:* Is the following transition matrix irreducible and aperiodic?

$$\boldsymbol{P} = \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{3} & 0 & 0 & \frac{2}{3} \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

The fine print

Definition: $f_{ii}^{(n)} = P(X_n = i, X_1 \neq i, \dots, X_{n-1} \neq i | X_0 = i)$ is the probability of first recurrence to *i* at the *n*th step.

Then $f_i = f_{ii} = \sum_{n=1}^{\infty} f_{ii}^{(n)}$ is the probability of recurrence.

Definition: A state *i* is called recurrent if $f_i = 1$, that is, eventual return is certain.

Otherwise, if $f_i < 1$ the state is called *transient.*

Definition: If the mean time for recurrence is finite, the state is called positive recurrent, otherwise it is null-recurrent.

Note: For infinite-state Markov chains, all 12 combinations of (irreducible or not), (aperiodic or not), and (positive recurrent or null recurrent or transient) are possible.

Theorem: If a general Markov chain is positive recurrent, then there exists a unique finite invariant measure.

Finding the stationary distribution

- Using a computer, you can apply "brute force", meaning compute some very high power of $\boldsymbol{P}$ and then each row should be approximately equal to $\boldsymbol{\pi}$

- Eigen decomposition: Notice that $\boldsymbol{\pi}'\boldsymbol{P} = \boldsymbol{\pi}'$ implies that $\boldsymbol{\pi}$ is a left eigenvector of $\boldsymbol{P}$, corresponding to eigenvalue 1. Therefore, if you can find the eigenvalues of $\boldsymbol{P}$ it means you can find $\boldsymbol{\pi}$

- You can solve $\boldsymbol{\pi}'\boldsymbol{P} = \boldsymbol{\pi}'$ as a system of linear equations subject to $\sum \pi_i = 1$.

Rate of convergence

Theorem: The eigenvalues $\lambda_1, \dots, \lambda_s$ of a Markov transition matrix P satisfy the inequality $|\lambda_i| \leq 1, \forall\, i = 1, \dots, s$.
Theorem: All Markov chain transition matrices have at least one eigenvalue equal to 1.

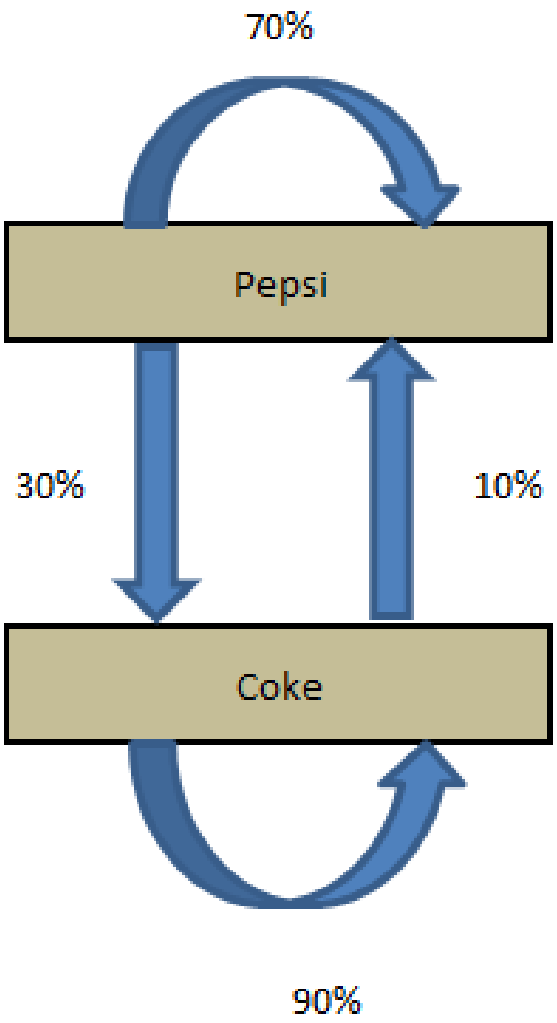
If all eigenvalues are positive and $\lambda_1 > \cdots > \lambda_s$, then $\lambda_1 = 1$ and the size of the next eigenvalue λ_2 indicates the rate of the speed of convergence as we approach equilibrium. The reason is because it describes how quickly the largest of the vanishing terms will approach zero in:

$$\boldsymbol{P}^n = \sum_{i=1}^{s} \lambda_i^n \boldsymbol{r}_i \boldsymbol{l}_i{}'$$

where r_i are the right eigenvectors, and l_i are the left eigenvectors.
Note: There is no formula how to compute the left eigenvectors if you already computed the right eigenvectors.

Business Example

Coke and Pepsi are the only companies in country X. A soda company wants to tie up with one of these competitor. They hire a market research company to find which of the brand will have a higher market share after 1 month and after 2 months. Currently, Pepsi owns 55% and Coke owns 45% of market share. Following are the conclusions drawn out by the market research company:



Likelihood Inference

Assume we have observed data $s_0, s_1, \ldots, s_n$ at times $0, 1, \ldots, n$ from a stationary Markov chain X_t. Given tjat sequence, we can estimate the transition matrix using the MLE method.

Then the likelihood is:

$$L(\boldsymbol{P}) = P\left(X_0 = s_0, \ldots, X_{t_n} = s_n\right)$$

$$= P(X_0 = s_0) \prod_{i=0}^{n-1} P\left(X_{t_{i+1}} = s_{t_{i+1}} \middle| X_{t_i} = s_i\right)$$

$$= P(X_0 = s_0) \prod_{i=0}^{n-1} p_{s_i s_{i+1}} = p_0 \prod_{i=1}^{s} \prod_{j=1}^{s} p_{ij}^{n_{ij}}$$

where n_{ij} is the observed frequency of transitions from i to j.

The loglikelihood is:

$$l(\boldsymbol{P}) = \sum_{i=1}^{s} \sum_{j=1}^{s} n_{ij} \log p_{ij} + \log p_0$$

MLE

If you take the derivatives,

$$\frac{\partial l(\boldsymbol{P})}{\partial p_{ij}} = \frac{n_{ij}}{p_{ij}}$$

It can be shown that the MLE of p_{ij} is

$$\hat{p}_{ij} = \frac{n_{ij}}{n_{i.}}$$

where $n_{i.} = \sum_{j=1}^{s} n_{ij}$

Note: there is a chi-square test if the transition probabilities are independent of the current state i.e. $p_{ij} = p_j$. It can be used to test if a data sequence is independent or a Markov chain or also to test convergence to equilibrium.

Extensions

One can extend the idea of a first-order Markov chain to chains of order m, where the probability of transition into s depends on the m previous states

$$P\left(X_j = s | X_0 = s_0, \dots, X_{t_{j-1}} = s_{j-1}\right)$$
$$= P\left(X_j = s | X_{j-m} = s_{j-m}, \dots, X_{t_{j-1}} = s_{j-1}\right)$$

When $m = 1$ we have $s(s-1)$ parameters There are S^m such vectors now!

Extensions

The state space S can in countably infinite. Then the transition "matrix" is infinite as well! The theoretical results are a lot more complicated.

A famous example is known as *random walk*.

For example, a random walk on the integer numbers $\mathbb{Z}$ can be defined as being at position z and changing to either $z + 1$ or $z - 1$ with probabilities

$$p_{z,z-1} = \frac{1}{2} + \frac{1}{2}\left(\frac{z}{c + |z|}\right)$$

$$p_{z,z+1} = 1 - p_{z,z-1}, z \in \mathbb{Z}$$

where $c > 0$.

Note that the set of all these p form an infinite matrix where most of the elements are zeros.

Road ahead

- Markov chain theory is most widely used in the *Markov chain Monte Carlo* (MCMC) methods in statistics.
- However, in MCMC the state space is usually *uncountable*, for example, $S = (-\infty, \infty)$, that is, $X_n \in S$ would be a continuous random variable.
- The transition matrix becomes a transition *kernel*

(A matrix can be regarded as an operator from vectors in $\mathbb{R}^s$ to $\mathbb{R}^s$, whereas the kernel transforms one density function to another density function)

- The equilibrium distribution is not a vector but a pdf.
- The conditions for ergodicity are more complicated
- The eigenvectors are called eigenfunctions.
- There are infinitely many eigenvalues!

Example

- Consider (again) a random walk defined as

$$X_n = \sum_{i=0}^{n} Z_i$$

where $Z_i \sim N(0,1)$ are independent.

Then the transition kernel is a conditional density given by

$$k(\cdot\,|x) = N(x,1)$$

This should be interpreted as the conditional density of X_{n+1} given that $X_n = x$.

What does this mean?

$$P(a < X_{n+1} < b | X_n = x) = \int_a^b \frac{1}{\sqrt{2\pi}} e^{-\frac{(y-x)^2}{2}} dy$$

Note that the Lebesgue measure is the invariant measure in this case.